



ML and FinTech

Huei-Wen Teng

Department of Information Management and Finance
National Yang Ming Chiao Tung University

Outline

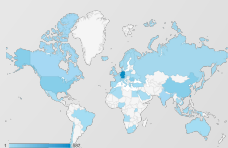
1. About ME: <https://venteng.github.io>

2. Syllabus

- ▶ The course centers on **data analytics** within FinTech!
- ▶ [Google form](#) for an quick update in course planning
- ▶ Infrastructure
- ▶ Cold calls during class

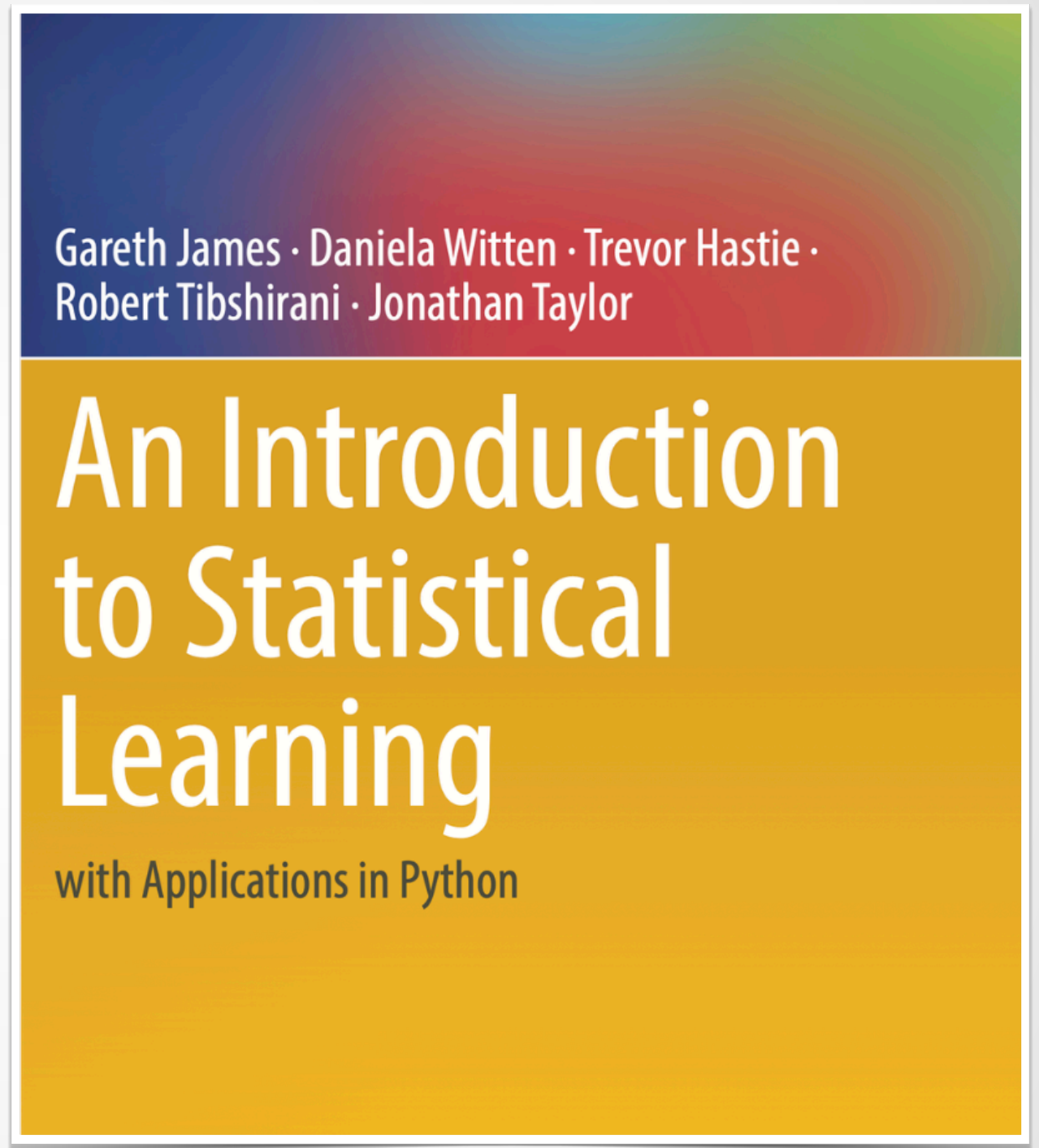


Category	Index	Type
participation (20%)	1	In-class exercise: with your laptop, or handwriting sheets (13%)
	2	presentaitons: MFS, R1, R2, R3, R4 (see invidiual project) (7%)
Exam (50%)	1	One A4 formula sheet is allowed in the exam.
Individual project (30%)		Individual, Replicating a high-quality paper. Need to provide slides and manuscript.
	R1	Topic: Paper introduction, Motivation of this paper, and how it connects to your peresonal interests
	R2	Data: Raw data, descriptions, EDA, and more
	R3	Benchmark model and experiement design (ablation studies)
	R4	Empirical analysis and conclusion



Introduction

- ▣ MUST:
 - ▣ Laptop & internet: In-class exercises and presentations
- ▣ TA: 陳之怡 (Claire Chen)
Email:
claire.mg14@nycu.edu.tw



<https://www.statlearning.com/>



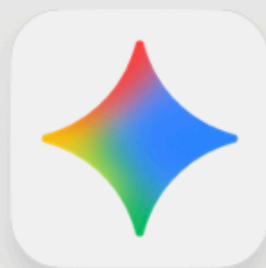
We embrace genAI!



New Chat



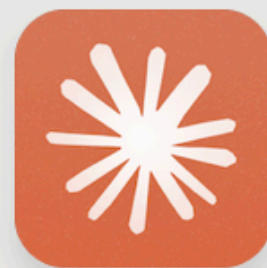
Grok



Gemini



Perplexity



Claude



國立陽明交通大學圖書館

National Yang Ming Chiao Tung University Library

服務

學習 · 研究

資源

校史 · 特藏

關於

AI研究助理

查館藏 ALMA

查引文 WOS

深度引文分析
Scite.AI



Infrastructure

- ▣ **E3:** Scores recording, announcement
- ▣ GitHub
 - ▶ An new course organization: 202609-ML-FinTech
 - ▶ Course info: 00-course-info
 - ▶ Your private repo: <studentID>-<nickname>
- ▣ Manuscripts in a professional writing style using latex
- ▣ Link to collect info for GitHub, overleaf



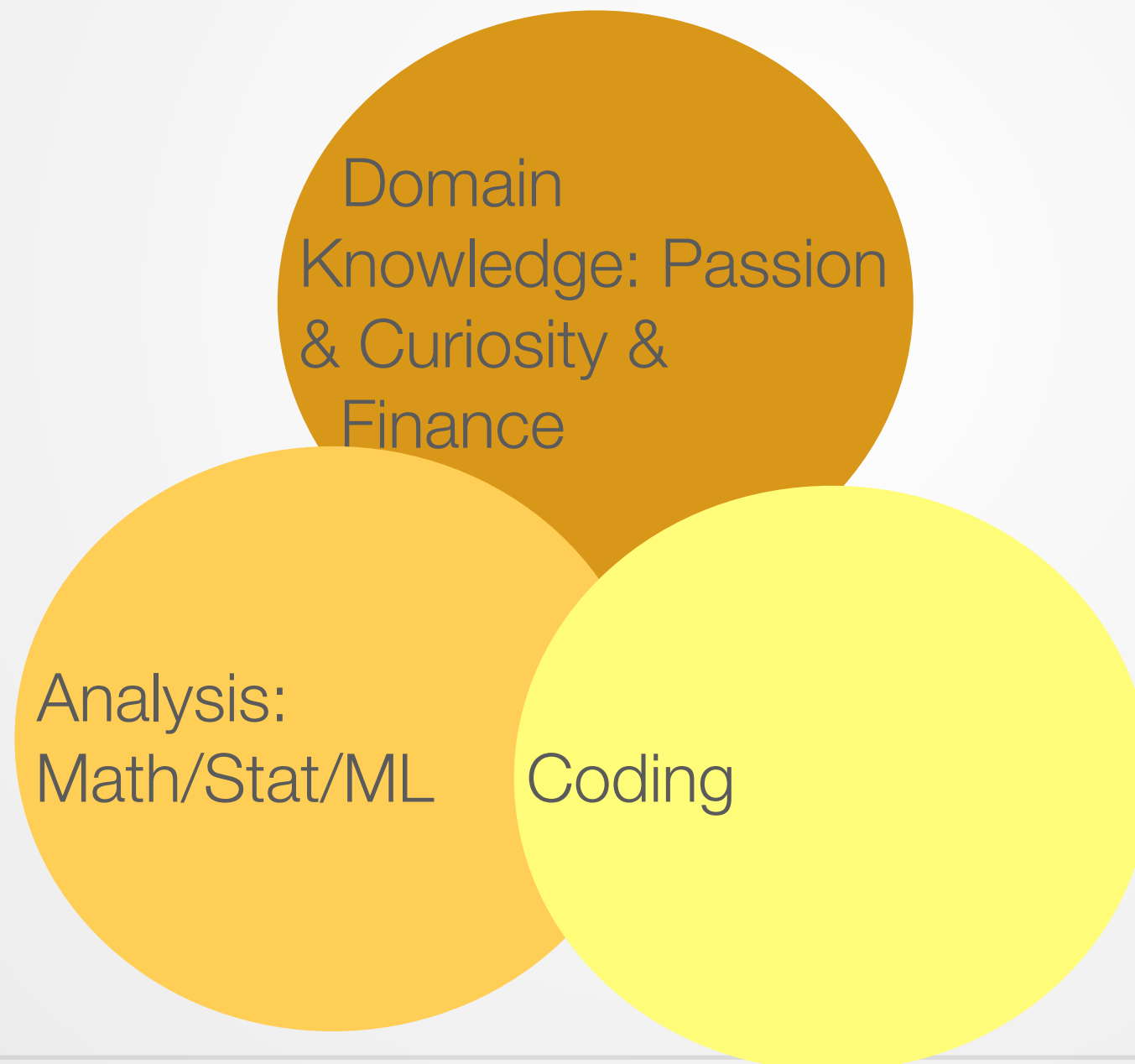
Topics

- ▣ Techniques
 - ▶ Machine Learning (supervised and/or unsupervised learning)
 - ▶ LLM
- ▣ Topics
 - ▶ Credit scoring/credit rating
 - ▶ Fraud detection
 - ▶ Trading strategies
 - ▶ Portfolio management
 - ▶ Others related to FinTech



Presentations, presentations, presentations!

- ▣ Slides: a template in Canva (coming soon)
- ▣ Manuscript: latex editing in Overleaf (coming soon)



Your GitHub Repo and an example

- ▣ README.md:
 - ▣ Link to your own GitHub page
 - ▣ Replicating a paper: Chicago style to reference a paper
 - ▣ Links to your manuscript via overleaf
 - ▣ Links to Slides via Canva
- ▣ Slides
- ▣ “coding”: Python in Jupyter Notebook
- ▣ “rawdata”: datasets or links to data (in README.md).



Potentials of your projects

- ▣ With modifications (new dataset, or new methods)
 - ▣ Thesis or dissertation
 - ▣ Journal publication



Tools

- ▣ Markdown使用說明
 - ▣ <https://hackmd.io/8nPFj8X7Rc2UhkhfjYAbkw?both>
- ▣ 哪裡可以找到資料?
 - ▣ <https://hackmd.io/LfakJmiPQCauby48zAx71xw>
- ▣ 高速運算: 國網中心
 - ▣ <https://hackmd.io/HXY75BRpRzimkWdCIAbuLw?both>
- ▣ A data competition hold at Stat, Sinica
 - ▣ <https://www3.stat.sinica.edu.tw/pds2026/>





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